FinSentry

Financial Transaction Anomaly & AML Detection — Project Report

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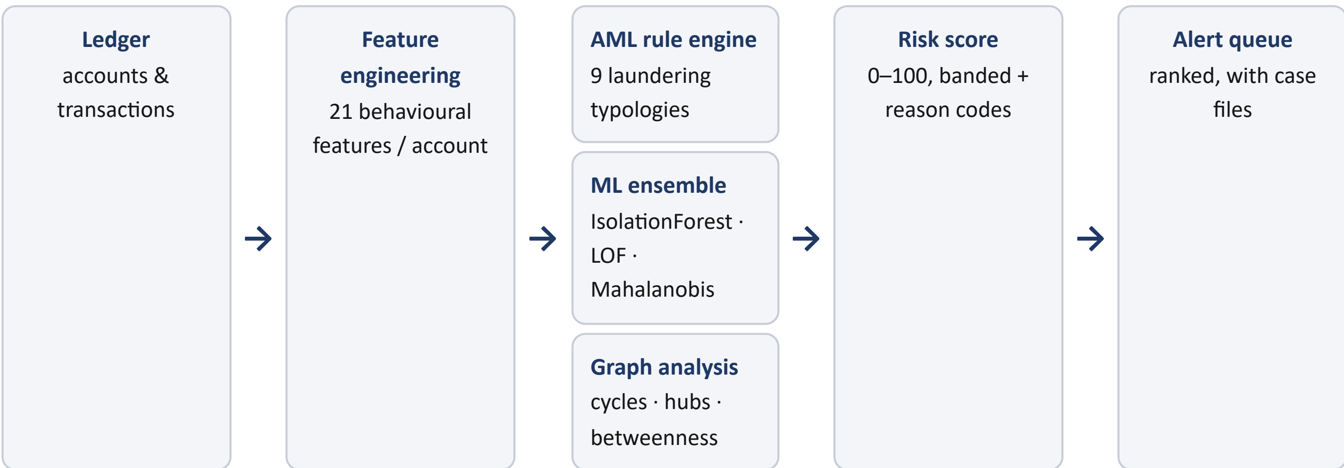
FinSentry is a working **transaction-monitoring (TM) system** that scores every account in a financial ledger for money-laundering risk. It mirrors how real bank AML platforms operate: instead of relying on a single model, it blends three independent signals — a deterministic **AML typology rule engine**, an unsupervised **machine-learning anomaly ensemble**, and **money-flow graph analysis** — into one explainable **0–100 risk score**, ranked into an analyst alert queue with human-readable reason codes.

0.96	0.93	0.90	9	3
ROC-AUC	PRECISION	RECALL	AML TYPOLOGIES	DETECTION ENGINES

1. Problem & objective

Banks are legally required to monitor transactions for signs of money laundering and file reports (STR/CTR) on suspicious activity. Production TM platforms (e.g. Actimize, SAS AML) combine rules, statistical/ML detection and link analysis, and every alert must be **explainable** to an investigator. FinSentry is a compact, end-to-end implementation of that exact pattern — built on synthetic data with injected laundering typologies and ground-truth labels, so detection quality can be measured objectively.

2. System architecture



Each account is reduced to a behavioural feature vector, scored by all three engines in parallel, and the signals are blended into a single risk score that feeds the alert queue.

3. Detection methods

3.1 AML typology rule engine

Deterministic detectors for the classic laundering patterns. Each fires a weighted, plain-English reason code.

TYPOLOGY	WHAT IT DETECTS
Structuring	Multiple cash deposits just below the reporting threshold
Smurfing	Many small inbound transfers from many distinct senders into one collector
Rapid movement / layering	Large inflow forwarded out again within hours (pass-through)
Circular flow / round-tripping	Funds cycle A→B→C→A through a small set of parties
High velocity	Transaction rate far above the account's population norm
Cash-intensive	High cash share over large throughput
Cross-border high-risk	Activity touching high-risk jurisdictions
Round amounts / odd hours	Unusual round-number activity or off-hours bursts

3.2 Unsupervised ML anomaly ensemble

Three complementary unsupervised models over standardized account features, averaged into one anomaly score, with a per-account explanation of the features that deviate most from the population:

- **Isolation Forest** — isolates rare, anomalous accounts via random partitioning.
- **Local Outlier Factor** — flags accounts whose local density differs from their neighbours.
- **Robust Mahalanobis distance** — multivariate distance from a robust covariance estimate.

3.3 Money-flow graph analysis

A directed value graph is built over material (high-value) flows and analysed with NetworkX for structural laundering signals: **circular flows** (round-tripping / layering), **fan-in / fan-out hubs** (collector / mule patterns), and **betweenness centrality** (pass-through intermediaries).

3.4 Composite risk scoring

The three signals are blended (default weights: rules 0.5, ML 0.3, graph 0.2) into a **0–100 risk score**, banded `LOW` `MEDIUM` `HIGH` `CRITICAL` , with the contributing reason codes attached to every alert. Weights are adjustable live in the dashboard.

4. Results

Evaluated against injected ground-truth labels on the synthetic generator (600 accounts, ~8,400 transactions, 6% suspicious):

ALERT BAND	PRECISION	RECALL	F1	ROC-AUC
MEDIUM+ (full alert queue)	0.93	0.90	0.91	0.96
HIGH / CRITICAL (high-confidence)	1.00	0.41	0.59	0.96

The high-confidence tier raises only the most certain cases (100% precision) for priority review, while the full queue catches ~90% of suspicious accounts — mirroring how a TM team triages by severity. Reproducible via `python evaluate.py`.

5. The analyst dashboard

An interactive console (live on Vercel; full version as a Streamlit app) lets an analyst work the output end-to-end:

- **Overview** — risk distribution, accounts by band, typologies triggered, flagged value over time.
- **Alert queue** — sortable, filterable, exportable list of accounts at MEDIUM risk or above.
- **Case detail** — risk gauge, reason codes, key features, transaction timeline, SAR/STR-style JSON export.
- **Network** — money-flow graph with alerts highlighted and detected circular flows listed.
- **Evaluation** — ROC curve, confusion matrix and per-model score distributions vs ground truth.

6. Technology & relevance

Stack: Python · scikit-learn · NetworkX · SciPy · pandas · Plotly · Streamlit · Docker; the live dashboard is deployed on Vercel. The project maps directly to the day-to-day of a transaction-monitoring analyst — reviewing alerts, recognising laundering typologies, documenting reason codes, and escalating high-risk cases.

Data & ethics: all data is synthetic, generated locally with reproducible seeds. No real customer or transaction information is used; thresholds and typologies are illustrative of real AML controls.